
COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB REPORT

Lab Title:

Development of Shop Inventory Management System Using C++

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1. OBJECTIVE

The objective of this lab is to develop a **Shop Inventory Management System** using the C++ programming language. The system is designed to maintain product records efficiently by allowing users to add, search, update, delete, and display products. It also performs stock value calculations, sorts products, and stores inventory records permanently using file handling. The project enhances programming skills in object-oriented programming, structures, arrays, functions, and file management.

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2. THEORY

Inventory management is an essential part of every business organization. It helps shopkeepers maintain records of products, monitor stock levels, update prices, and calculate the total value of inventory. Manual inventory management is time-consuming and prone to errors; therefore, computerized inventory systems provide a more efficient solution.

This project is developed using the **C++ programming language**, which supports object-oriented programming concepts such as classes, structures, encapsulation, and modular programming. A **Product** structure is used to store product information including Product ID, Product Name, Quantity, Price, and Total Value.

An **Inventory** class manages all operations related to inventory. Arrays are used to store multiple product records, while separate functions perform operations such as adding products, searching, updating quantity, updating price, deleting products, sorting records, and calculating stock value.

The program also implements **file handling** using input and output file streams. Inventory data is stored in a text file named **inventory.txt**, allowing users to save records permanently and reload them whenever the program starts.

The inventory system provides a simple menu-driven interface that makes inventory management efficient and user-friendly.

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3. SOFTWARE USED

- DEV-C++ IDE
- C++11 Compiler
- Windows Operating System
- Text File (**inventory.txt**) for permanent storage

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4. PROCEDURE

Step 1 – Create a New Project

- Open DEV-C++.
- Create a new C++ source file.
- Save the file as **SHOP INVENTRY MANAGEMENT SYSTEM.cpp**.

Step 2 – Include Required Header Files

Include the following libraries:

- iostream
- iomanip
- fstream
- string
- sstream

These libraries provide console input/output, table formatting, file handling, string manipulation, and file parsing.

Step 3 – Create Product Structure

Create a **Product** structure containing:

- Product ID
- Product Name
- Quantity
- Price
- Total Stock Value

Step 4 – Create Inventory Class

Create an **Inventory** class that stores product records and provides member functions for inventory management.

The class maintains an array of products and keeps track of the total number of products stored.

Step 5 – Implement Inventory Functions

Develop separate functions for the following operations:

- Add Product
- Display Products
- Search Product
- Increase Product Quantity
- Decrease Product Quantity
- Update Product Price
- Delete Product
- Calculate Total Stock Value
- Sort Products by ID
- Save Inventory Data
- Load Inventory Data

Step 6 – Calculate Product Value

Whenever a product is added or updated, calculate:

Total Value = Quantity × Price

The total stock value of all products is also calculated by summing the value of every product in the inventory.

Step 7 – Display Products

Display inventory records in a formatted table containing:

- Product ID
- Product Name
- Quantity
- Price
- Total Value

The **iomanip** library is used to align the table properly.

Step 8 – Save and Load Inventory

Store inventory records in the file:

inventory.txt

The system can reload all saved records whenever the application starts.

Step 9 – Compile and Execute

Compile the program.

Run the application.

The following menu appears:

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  Shop Inventory Management System
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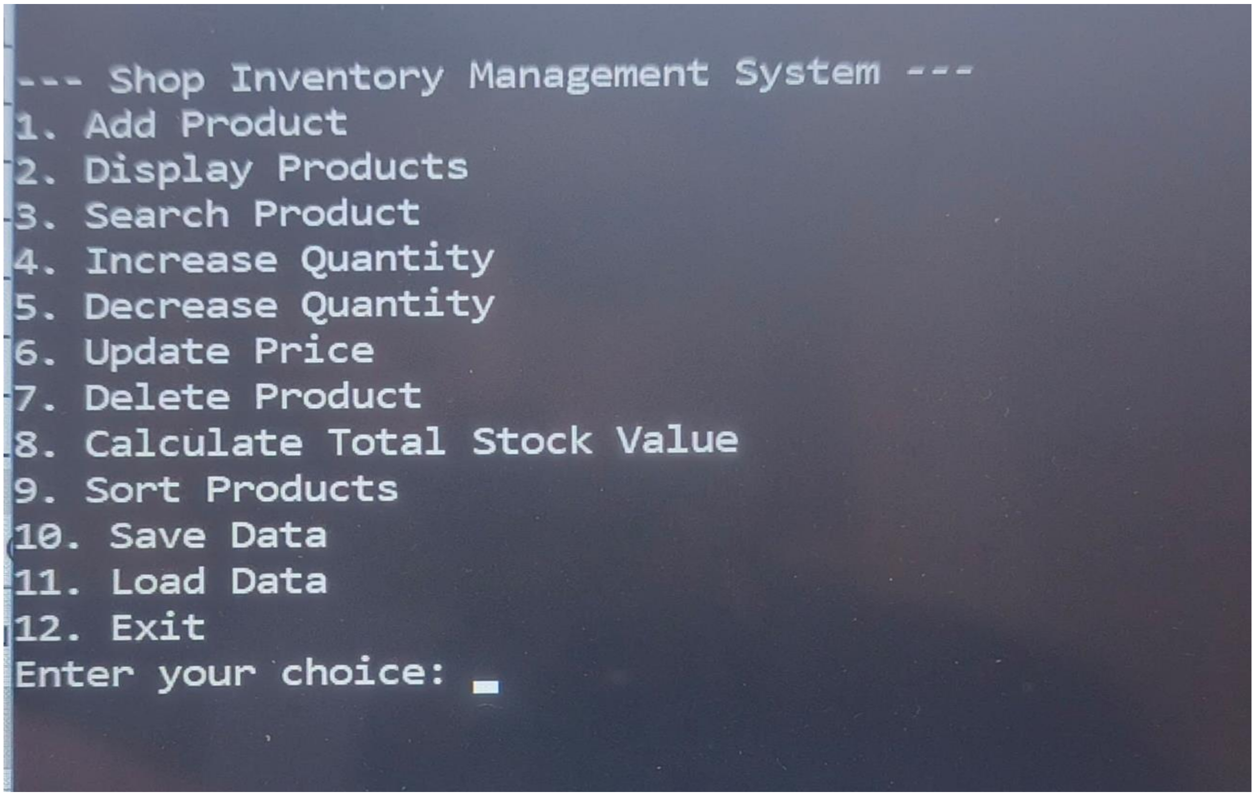

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1. Add Product

2. Display Products

3. Search Product

4. Increase Quantity

5. Decrease Quantity

6. Update Price

7. Delete Product

8. Calculate Total Stock Value

9. Sort Products

10. Save Data

11. Load Data

12. Exit

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5. PROGRAM FEATURES

The developed inventory system provides the following features:

- Add new products to inventory.
- Display all products in a tabular format.
- Search products using Product ID.
- Increase available stock quantity.
- Decrease available stock quantity.
- Update product prices.
- Delete products from inventory.
- Automatically calculate total value of each product.
- Calculate total stock value of the shop.
- Sort products according to Product ID.
- Save inventory records using file handling.
- Load saved inventory records automatically.
- User-friendly menu-driven interface.

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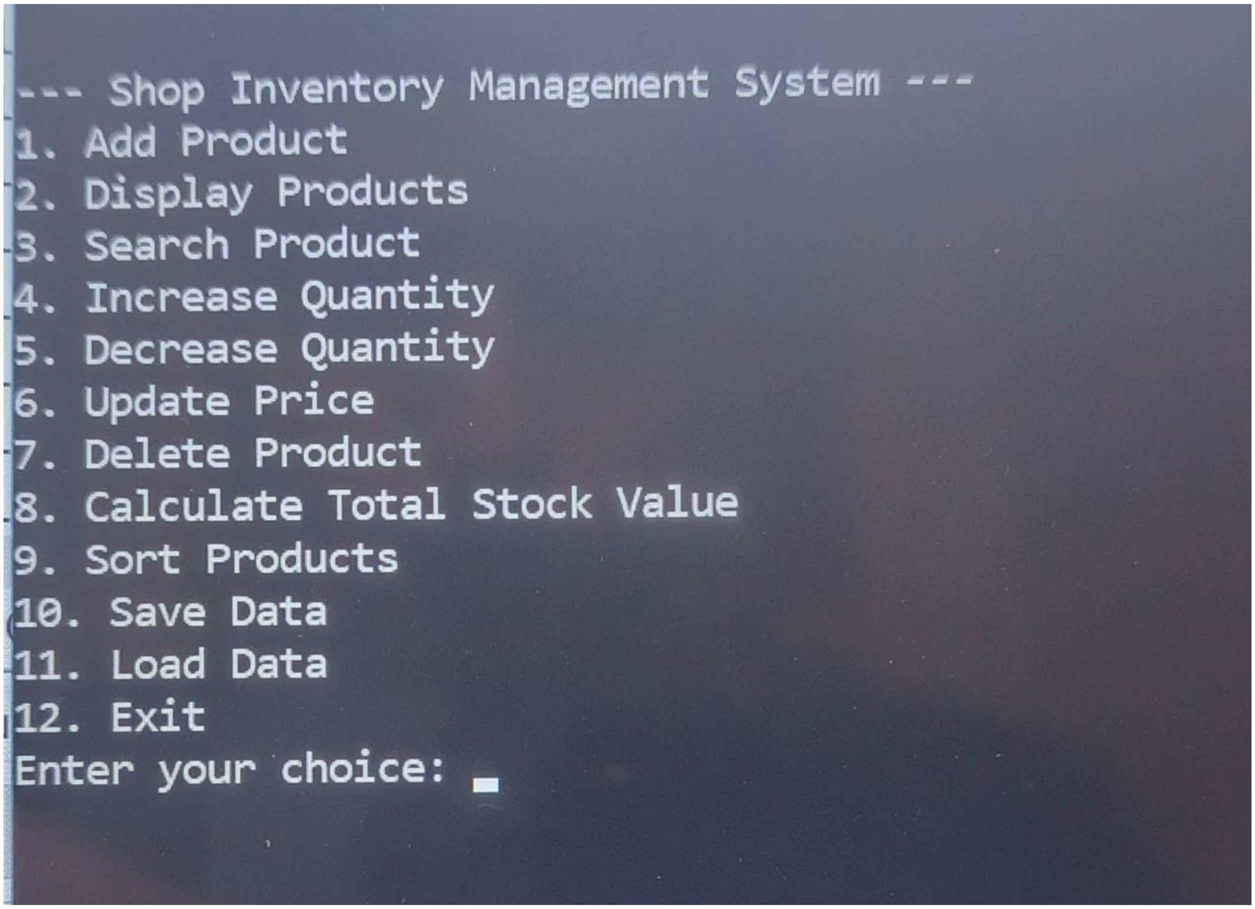
6. SAMPLE OUTPUT

Main Menu

Shop Inventory Management System



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1. Add Product

2. Display Products

3. Search Product

4. Increase Quantity

5. Decrease Quantity

6. Update Price

7. Delete Product

8. Calculate Total Stock Value

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12. Exit

Adding a Product

Enter Product ID : 101

Enter Product Name : Laptop

Enter Quantity : 15

Enter Price : 65000

Product added successfully!

Display Products

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ID	Name	Quantity	Price	Total Value
1	tomatoes	10	150.00	1500.00
2	rice	100	500.00	50000.00
3	mangoes	200	250.00	50000.00
4	banana	300	200.00	60000.00
5	apples	400	100.00	40000.00

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Total Stock Value

Total Stock Value = 201500

Search Product

Enter Product ID to search : 2

Product Found

ID	Name	Quantity	Price	Total Value
2	rice	500	500.00	50000.00

Save Data

Data saved to file successfully.

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7. CONCLUSION

The Shop Inventory Management System was successfully developed using the C++ programming language. The project demonstrated practical implementation of object-oriented programming concepts, structures, arrays, functions, sorting algorithms, formatted table output, and file handling. The system efficiently manages inventory records by allowing users to add, update, search, delete, and display products while automatically calculating stock values. Permanent data storage using text files makes the system reliable for future use. Overall, this project provided valuable experience in developing a real-world inventory management application and strengthened programming skills in C++.